# Midterm - Fall 2026

*ECON 5783 — University of Arkansas*

1. Say a researcher wants to model the conditional expectation of annual earnings given gender, state of residence, and whether a worker has a college degree. They have a sample of 1,000 workers. Gender is recorded male and female, state of residence (50 states), and college degree is a binary indicator.
   (a) Describe how you would estimate this conditional expectation function "fully flexibly". Use college-educated women in Arkansas as an example.
   (b) Why might this approach be impractical with this data?
   (c) The researcher instead uses a linear model with an intercept, an indicator for being male, an indicator for having a college degree, and indicators for each of the 50 states (omitting a reference category for each variable). What does this model leave out relative to the fully flexible CEF? Give an example of a pattern in average earnings that this model would not be able to capture.

2. A researcher wants to estimate the returns to college using men in the National Longitudinal Survey of Young Men. The outcome is hourly wages in 1976, and treatment is completing at least 16 years of schooling. The table below compares average characteristics of men who attended and did not attend college (before any matching). The characteristics describe family background and where the men lived in 1966.

| | High School | College | Difference |
|---|---|---|---|
| Mother's years of schooling | 9.98 | 12.03 | 2.15 |
| Father's years of schooling | 9.05 | 11.94 | 2.89 |
| Lived with both parents at age 14 | 0.933 | 0.967 | 0.034 |
| Lived in the South in 1966 | 0.359 | 0.313 | −0.046 |
| Number of observations | 685 | 703 | |

(a) What does this table tell you about the comparability of the two groups? Discuss at least two differences and explain which direction they would likely bias the difference-in-means estimate.

(b) The researcher decides to use nearest-neighbor matching on the four variables in the table. Describe in words the nearest-neighbor matching procedure. You do not need to write a formula or code; just describe in words the steps that one would do.

(c) Instead of matching, the researcher might want to use a regression adjustment approach. Describe in words the steps to implement this estimator. You do not need to write a formula or code; just describe in words the steps that one would do.

The following questions refer to "The Fox News Effect: Media Bias and Voting" by DellaVigna and Kaplan (2007). The authors study whether making Fox News available on local cable affects voting. By the 2000 presidential election, some towns had access to Fox News while others did not.

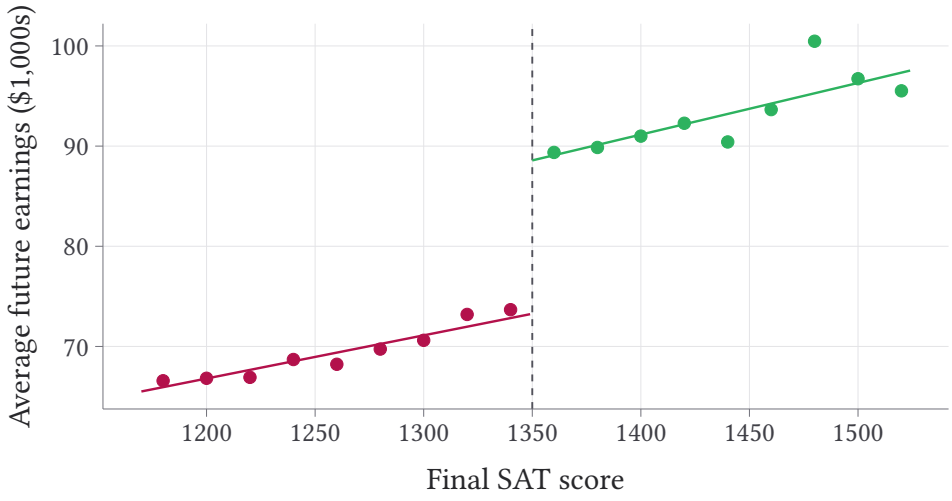
3. Say a researcher compares the average Republican vote share in 2000 between towns with and without Fox News. Do you think this difference-in-means estimate identifies the causal effect of Fox News availability? Give a reason why the estimate might be biased, predict the sign of the bias, and explain your thinking.

4. The authors present evidence that, conditional on geographic and cable-market controls, Republican vote share in 1996 did not predict whether a town had Fox News by 2000. Why might this be useful information for assessing the plausibility of the seleciton on observables assumption? Explain your thinking.

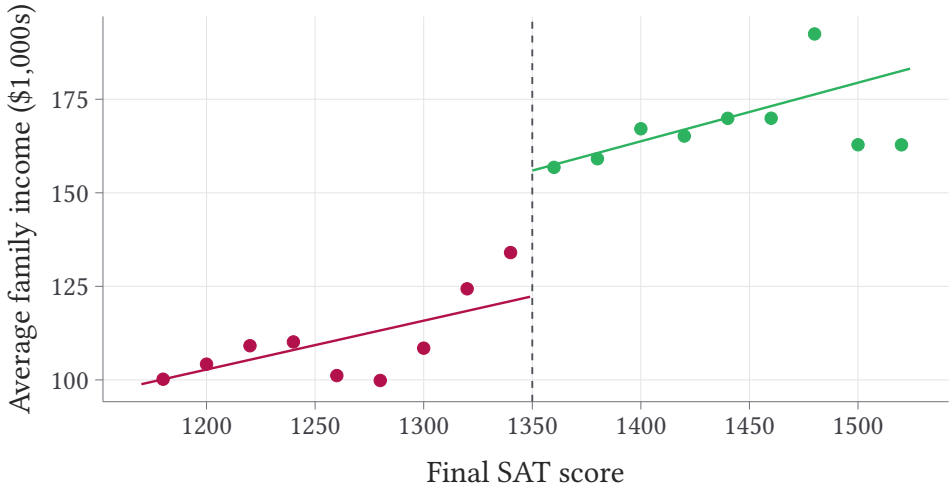
For the following questions refer to Figure 1. These are two figures created using synthetic data. The setup is that students in Michigan are guaranteed admission to the University of Michigan if they get at least a 1350 on their SAT. Students, of course, are allowed to retake the SAT as many times as they want and the top score is used. A researcher wants to estimate the effect of guaranteed admission on students' future earnings.

5. Figure 1(a) presents the researcher's main result. What do we need to assume for the jump in future earnings at 1350 to identify the causal effect of guaranteed admission?
6. Figure 1(b) shows average family income, measured before students take the SAT. Do you think this figure strengthens or weakens the researcher's causal claim? Explain how it relates to the assumption you discussed in the previous question.
7. The researcher responds by using only students with SAT scores very close to 1350. Do you think using a smaller bandwidth fixes the concern you described above? Why or why not?

**Figure 1 — Simulated Data**




**(a) RDD Estimate on Student's Future Earnings**




**(b) RDD Estimate on Family Income**